Farihal Abedin

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Named Contributor, IEEE 2418.5–2025: Guide for Blockchain in Power and Energy Systems
World Bank Government Anlaytics Fellow, 2026

Research Interests

Smart and distributed infrastructure, Human AI interaction, Technology development for real-world deployment

Education

Bangladesh University of Engineering and Technology (BUET) 2024
M.Sc., Electrical and Electronic Engineering | CGPA 3.75/4.00 Dhaka, Bangladesh

Bangladesh University of Engineering and Technology (BUET) 2017
B.Sc., Electrical and Electronic Engineering | CGPA 3.69/4.00 | class rank 66 Dhaka, Bangladesh

Selected Research Experience

M.Sc. Thesis: PMSM as replacement for single-cylinder diesel boat engines for rural water transportation 2023 – 2024
First author; published in Next Research, 2026 Supervisor: Dr. Md. Ziaur Rahman Khan

- Designed a 12 kW, 2200 RPM interior PMSM (48-slot, 8-pole, V-shaped NdFeB magnets, water-jacket cooling) in ANSYS Motor-CAD to match the operating point of legacy diesel engines used in rural inland waterways.
- Ran electromagnetic, mechanical and thermal analyses; a three-step rotor skewing cut cogging torque by 59% ; peak torque 1.3% of rated torque, torque ripple 7.32%.
- Achieved >93% efficiency at rated and peak torque (target 92%), 52.10 N·m at 2200 RPM, and a rotor safety factor of 7.85 at 110% of maximum speed.
- Evaluated over the ISO 8178-4 type E5 duty cycle: 93.81% energy utilization - a far greater value compared to the engines it replaces.

MPPT with machine learning for solar modules 2022
Co-author; published in Frontiers in Energy Research, 2022 Supervisor: Late Dr. Kazi Mujibur Rahman

- Built a PV model in MATLAB/Simulink and generated 1,300 simulations across temperature and irradiance (datasets released publicly); trained a Bayesian-regularized neural network (15 hidden neurons) to predict the maximum power point.
- Reached >95% tracking efficiency versus 67.4% (P&O), 81.39% (IC) and 90.08% (fuzzy logic), with ∼1 ms inference and robustness to noisy inputs.
- Validated on a low-cost Arduino, boost-converter and solar-panel testbed.
- My role: simulation, dataset generation.

R3Arm: Gesture-controlled robotic arm for rescue operations 2017
Co-author; IEEE R10-HTC 2017

- Built a low-cost, modular 3-DOF robotic arm on a mobile base, controlled by finger-mounted accelerometer gestures over an nRF24L01 link, with live smartphone video for remote operation.
- My role: gesture acquisition, firmware, mechanical design.

R3Diver: Low-cost underwater rescue ROV 2016
Co-author; IEEE TENCON 2016 Supervisor: Dr. Shaikh Anowarul Fattah

- Built a USD 100 remotely operated vehicle (PVC frame, four thrusters, 3D-printed propellers, 12 motions) with wired USB-over-CAT6 control and live video to 30 m.
- Integrated automatic human detection in the video feed using HOG and Haar cascade classifiers.
- My role: control, thruster design, HMI, vision.

IEEE P2418.5 Working Group (Blockchain in Power and Energy Systems) 2021 – 2025
Named Contributor; standard published December 2025 IEEE

- Contributed to discussions on blockchain architecture functional layers; reviewed use cases including peer-to-peer energy trading and grid data management.
- Reviewed interoperability between blockchain systems and SCADA infrastructure.

Selected Publications (Google Scholar)

Journal Articles

J2 **F. Abedin** and M. Z. R. Khan, "Design and Analysis of a Permanent Magnet Synchronous Motor for Sustainable Rural Water Transportation," *Next Research*, 2026, p. 101589 (link)

J1 R. Sharmin, S. S. Chowdhury, **F. Abedin**, and K. M. Rahman, "Implementation of an MPPT technique of a solar module with supervised machine learning," *Frontiers in Energy Research*, vol. 10, 2022, p. 932653 (link)

Conference Papers

C2 P. P. Sarker, **F. Abedin**, and F. N. Shimim, "R3Arm: Gesture controlled robotic arm for remote rescue operation," in *Proc. IEEE Region 10 Humanitarian Technology Conf. (R10-HTC)*, 2017, pp. 428–431 (link)

C1 S. A. Fattah, **F. Abedin**, M. N. Ansary, M. A. Rokib, N. Saha, and C. Shahnaz, "R3Diver: Remote robotic

rescue diver for rapid underwater search and rescue operation," in *Proc. IEEE TENCON*, 2016, pp. 3280–3283 (link)

Technical Standards

S1 *IEEE 2418.5-2025: IEEE Guide for Blockchain in Power and Energy Systems*, IEEE, Dec. 2025. Named Contributor, Working Group P2418.5 (2021–2025)

Industry Experience

ICT Division, Gas Transmission Company Ltd. (GTCL) Jan 2024 – Present
Deputy Manager Dhaka, Bangladesh

- Maintain ICT systems supporting national gas transmission operations, including infrastructure for pipeline monitoring and SCADA-related systems.
- Scale: 57 sites / 2177 km of pipeline supported

ICT Division, Gas Transmission Company Ltd. (GTCL) Dec 2018 – Jan 2024
Assistant Engineer Dhaka, Bangladesh

- Managed enterprise network systems and operational software infrastructure.

Teaching Experience

Daffodil International University Mar – Dec 2018
Lecturer (Contract), Dept. of EEE Dhaka, Bangladesh

- Taught nine courses across power systems, VLSI, digital electronics, and programming.
- Supervised undergraduate theses, projects, and fieldwork; served on the Curriculum Development Committee.

Brahmanpara Upazila Administration Mar – Apr 2022
Instructor, Robotics and Programming (High School), short-term contract Cumilla, Bangladesh

- Taught Arduino-based robotics and programming; mentored the demonstration of a PID-controlled line-following robot using IR sensors and motor drivers.

Professional Service

Peer Review

- OzCHI 2026, IEEE Transactions on Magnetics – Conferences [INTERMAG 2026], GMSARN International Journal (Scopus Q3), OzCHI 2025, IEEE VIS 2025 alt.vis

Leadership and Community

- Topic Chair, IEEE Energy Conversion Congress and Exposition (ECCE) 2025 [Track-B; Duties: organizing paper reviews and rendering a preliminary accept/reject decision]
- Developer and Maintainer, Study Materials Archive Portal, Dept. of EEE, BUET, 2024 – Present
- Membership Development Coordinator, IEEE Power & Energy Society, Bangladesh Chapter, 2020 – 2023
- Proctor for BUET undergraduate teams, IEEEXtreme 14.0, 2020

Honors and Awards

- **World Bank Government Analytics Fellowship**, 2026
- **Champion**, Undergraduate Level, Bangladesh Electronics Olympiad, 2015

Test Scores

IELTS Academic, overall Band 7.5 (L: 8.0, R: 9, W: 7.0, S: 6.5), British Council, Sep 2026

Skills

- **Simulation and Design:** ANSYS Motor-CAD (FEA), MATLAB/Simulink, OpenCV, LabVIEW, Arduino
- **Programming:** Python, C/C++, MATLAB

References

Dr. Md. Ziaur Rahman Khan
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Dr. Shaikh Anowarul Fattah
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